Issues — Open Workable Items

Revision: 8 **Last modified:** 2026-08-19T18:07:03Z **Ticket prefix:** `BOB` (operator-mandated, 2026-06-06) **Scope:** Open/active items only. Closed items migrate to `Fixed.md`.

> Tracking: this file + `Issues_Summary.md` are authoritative for open work. The SQLite single-source-of-truth + `docs_chain` engine (BOB-010) is complete.

BOB-008 — RuTracker automated login blocked by CAPTCHA

Status: Operator-blocked **Type:** Bug **Created:** 2026-06-06 **Operator-Block-Details:** WHAT — RuTracker login with stored creds returns no session cookie (CAPTCHA wall). WHY — automated user/pass login is CAPTCHA-gated; self-resolution exhausted (creds correct + wired, login attempted, `auth=True`). UNBLOCK — [A] operator completes the CAPTCHA flow at `/api/v1/auth/rutracker/captcha` + `/login`. [B] operator pastes a fresh `bb_session` cookie via `/auth/rutracker/cookie-login`. WHO — operator.

Evidence: live search per-tracker stat `rutracker status=error auth=True error="login returned no session cookie — likely CAPTCHA"`.

BOB-065 — Lava P2: Egress diagnosis and VPN-host SOCKS routing (containers pkg/egress)

Status: Queued **Type:** Task **Severity:** High

[Backfill from RD2-15/GA-05, audit doc 2026-08-08] Lava-porting finding P2 (Egress decision + VPN-host routing, Lava PLAYBOOK sections 0 and 4). Problem Boba-Base shares: on a datacenter host, trackers are network-blocked (DNS-fail/TLS-MITM, not Cloudflare so FlareSolverr cannot fix). Affects Jackett indexer fetches + merge_service/download-proxy + plugin engines.

Diagnosis (port the script): curl https://api.ipify.org (host IP) + curl -o /dev/null -w %{http_code} https:/// direct vs via a VPN-host SOCKS proxy. Different egress IP + 200 via proxy confirms. Fix: route outbound through a VPN-connected host (the nezha pattern). SOCKS tunnel ssh -D 127.0.0.1:1080 -N (use –socks5-hostname for remote DNS); point Jackett + download-proxy + qBitTorrent-go at it (P3). For browser-cookie harvest, run the harvester ON the VPN host. Port: containers submodule pkg/egress (tunnel up/verify) + scripts/egress-via-vpn.sh glue; reuse Boba-Base existing ensure-macos-tunnel.sh style. TDD: assert the via-proxy egress IP != direct host IP AND a known-blocked tracker returns 200 via proxy. Source: docs/PORTING-FROM-LAVA.md. Per audit RD2-15 [P0]: Create tracked workable items (BOB-064..067) for the four Lava-porting findings, citing implementing commits as evidence, closed as Implemented.

BOB-066 — Lava P3: BOBA_UPSTREAM_PROXY in download-proxy + qBitTorrent-go + Jackett + compose env-forward

Status: In progress **Type:** Task **Severity:** High

[Backfill from RD2-15/GA-05, audit doc 2026-08-08] Lava-porting finding P3 (Configurable outbound proxy in the services, Lava PLAYBOOK section 3). download-proxy (Python): httpx/requests honor HTTP_PROXY/HTTPS_PROXY/ALL_PROXY/NO_PROXY env natively — add an explicit BOBA_UPSTREAM_PROXY config that sets these for tracker-bound clients, with loopback bypass (NO_PROXY=127.0.0.1,localhost,jackett). qBitTorrent-go: set http.Transport.Proxy (socks5 native, remote DNS) from a BOBA_UPSTREAM_PROXY env — mirror Lava internal/httpx/proxy.go. Jackett: has a built-in proxy setting (configure via its API/ServerConfig). Deploy gotcha (port): the env must be FORWARDED into the containers (docker-compose.yml env / the boba-ctl deploy) — Lava bug was a missing allow-list entry. Verify on distroless via podman inspect, not exec printenv. TDD: a test with a local proxy asserts the service tracker request traverses it; falsifiability: disable the wiring so the test fails. Source: docs/PORTING-FROM-LAVA.md. Per audit RD2-15 [P0]: Create tracked workable items (BOB-064..067) for the four Lava-porting findings, citing implementing commits as evidence, closed as Implemented.

BOB-068 — RD2-00: unattributed, unreviewed Auto-commit mechanism pushing to main

Status: Queued **Type:** Bug **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-00, Rev-6 downgrade to P1] Three commits at HEAD carry the bare message Auto-commit with no body, no ticket reference, no TDD trail: 54e313f (26 files), 9c8f684 (6 files), 743097a (2 files). Two landed while the audit was in progress. Source unidentified: no crontab, no systemd timer, no matching process, no in-tree script emitting the string, no hooksPath, no PostToolUse/Stop hook. Timezone +0500 on newer commits does not match host +0300 MSK. RD2-00 Update: git reflog proves 9c8f684/743097a arrived via pull: Fast-forward from a +0500 host, matching the established 2026-06-28 cross-host rsync-sync pattern (55b8671/cdb555f). Update 2 (Rev-6): root cause is a second live Claude session (Opus 5, same +0500 host) that independently landed constitution commit 177f2b0 (new anchor §11.4.238) while this session was mid-edit on the identical anchor number. Downgrades severity from P0 to P1. Every such commit bypasses §2 wrapper, §11.4.43 TDD-fix, §11.4.92 5-pass eval, §11.4.125/§11.4.142/§11.4.194/§11.4.209 mandatory Fable-xhigh review, §11.4.234 dedicated commit/push script. Composes with RD2-10..13.

BOB-069 — RD2-40: §11.4.238 QA-discovery-channel ledger — ongoing coverage-escape backfill

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md Rev-6 addendum, priority downgraded P0->P1] §11.4.238 compliance MECHANISM STOOD UP AND EXERCISED 2026-08-08. docs/QA_DISCOVERY_LEDGER.md created with discovery-channel schema, seeded with 4 real entries from this session (RD2-22, RD2-41a, RD2-41b, BOB-008), each with a real coverage-escape audit citing the specific missing/blind check; 3 of 4 already closed with a genuine new automated check carrying real RED->GREEN evidence. CLAUDE.md Critical Constraints now points at the ledger and states the discovery-channel-split mandate. Closed-loop example exercised end-to-end: scripts/pre_build_verification.sh invariant 17 extended to run workable-items diff (not just validate) — RED-captured via tests/unit/test_pre_build_workable_items_diff_check.sh §1.1 paired mutation. Honest boundary (§11.4.6): full retroactive audit of every one of the ~50 GA-NN/RD2-NN findings in this document (each needs its own escape-audit entry) was NOT attempted — ledger is honest that it starts now (100% out-of-band in its own tracked split table) rather than backfilling a false complete history. BOB-009/BOB-010 evidence_path gap remains open. Priority: downgraded from P0-blocking to P1-ongoing — MECHANISM exists and applies to new findings; the remaining retroactive audit + BOB-009/010 closure is incremental.

BOB-070 — RD2-41: pre-build mutation-marker scan carrier false-positive silently defeats entire pre-build gate

Status: Queued **Type:** Bug **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md Rev-6 addendum, RD2-41 P1 — arguably P0] scripts/docs_chain.sh (FIXED this session) and scripts/pre_build_verification.sh invariant 17 (FIXED this session) both hardcoded a non-existent bin/workable-items path, silently no-op ing/skipping DB-export and DB-validate steps on every run — root-caused and fixed via the resolution chain proven in constitution/scripts/reporting/report_item.sh (env override -> committed constitution copy -> on-demand go build). Regression guards: tests/unit/test_docs_chain_binary_resolution.sh (RED->GREEN) + tests/unit/test_pre_build_workable_items_invariant.sh (RED->GREEN). A SEPARATE, still-open defect surfaced while fixing this: pre_build_verification.sh pre-code-review mutation-marker scan aborts the ENTIRE script (exit 1) before invariant 17 is ever reached, on carrier false-positives — legitimate constitution/**/*_mutation_test.sh + *_test.go files that intentionally contain the literal strings MUTATED/# MUTATION as part of their OWN §1.1 testing logic (36 hits, all under constitution/scripts/{gates,multitrack,workable-items}/). Same §11.4.201 carrier-false-positive class as GA-24 + RD2-01. Practical impact: the ENTIRE pre-build gate has not completed a full run for as long as this false-positive has been live — no invariant past the mutation-marker check has been genuinely enforced. Fix needed: marker-scanner to exclude self-referential test/gate files (or use structural check than bare substring), matching the fix scoped for GA-24/RD2-01/RD2-36. Priority: P1, arguably P0.

BOB-074 — RD2-07: DDoS-class testing fully absent from the mandated test-type matrix

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-07, P2] tests/ has 18 subdirectories covering unit/integration/e2e/security/chaos/stress/performance/benchmark/UI (13 of the constitution ~14 mandated categories present). tests/load/locustfile.py covers throughput/scale only. No file anywhere mentions DDoS, flood, or attack-resilience testing — a distinct mandated category per §11.4.27, and not legitimately N/A given this is an internet-facing FastAPI/Gin service with public HTTP endpoints reachable over a LAN tunnel. Fix: author tests/ddos/ (or extend tests/load/) with rate-limit-enforcement, malformed-request-flood, and resource-exhaustion-under-attack coverage for the exposed download-proxy/merge-service endpoints. Priority: P2. Canonical implementation: RD2-32.

BOB-076 — RD2-09: submodules/jackett fork 1 commit behind upstream (informational)

Status: Queued **Type:** Task **Severity:** Low

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-09, P3 informational] git submodule status + fetch shows submodules/jackett 1 commit behind origin/master (e12342eb4, a trivial false-positive fix). All other submodules (constitution, challenges, containers, helixqa) are exactly at their upstream tip. Fix: bump when convenient; not blocking anything. Priority: P3. Canonical implementation: RD2-39.

BOB-077 — RD2-10: Identify second host running the Auto-commit rsync/sync mechanism (OPERATOR-DECISION)

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: OPERATOR must identify which second host holds push credentials + confirms whether the rsync/sync mechanism is intentional or a stale job to retire. This session cannot inspect a host it has no access to (§11.4.6/§11.4.101). WHY: This session cannot inspect or reach the +0500 host that produced the Auto-commit fast-forwards (§11.4.6 no-guessing, §11.4.101 reversible-safe default). UNBLOCK: [A] operator names the second host + confirms intentional (proceed to RD2-11 wiring) · [B] operator confirms stale/misconfigured (retire the job) · [C] operator confirms the second live Claude session/device is the source (Rev-6 Update 2 root cause, proceed to per-session discipline enforcement) WHO: Operator **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-10, P0 OPERATOR-DECISION] Root-caused as far as this host allows (see RD2-00 Update): reflog proves the two newest commits arrived via plain pull: Fast-forward from a +0500 host, matching this project established 2026-06-28 cross-host rsync-sync pattern (cdb555f/55b8671). Needs operator input to go further — which second host runs it, and is it the intended mechanism (just needs a real message + review gate) or a stale job to retire. Not auto-executed (§11.4.6/§11.4.101 — this session cannot inspect or guess at a host it has no access to). Rev-6 update: root cause narrowed to a second live Claude session (Opus 5, same +0500 host) that independently landed constitution 177f2b0. Priority: P0 (operator-blocked).

BOB-078 — RD2-11: Once identified, wire Auto-commit mechanism through §11.4.234 dedicated commit/push script OR retire it

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-11, P0] Once identified (RD2-10): either wire it through a real §11.4.234 dedicated commit/push script (with the hook-validation stages this project already has via .claude/settings.json PreToolUse guard, extended to a real pre-commit content check) or shut it down if it serves no purpose. Priority: P0. Blocked on RD2-10.

BOB-079 — RD2-12: Retroactive attributed history notes for GA-18/21/22/25/26/27 changes (never rewrite published history)

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-12, P1] Retroactively give proper, attributed, reviewed commit messages / history notes to the technically-correct changes the Auto-commit mechanism already made (GA-18, GA-21, GA-22, GA-25, GA-26, GA-27) — this can be a documentation/changelog note rather than a history rewrite (§11.4.113 — never rewrite published history), citing the git-history evidence per §11.4.124 investigate-before-attribute pattern. Priority: P1.

BOB-080 — RD2-13: Retroactive Fable-xhigh code review of the substantive Auto-commit diffs

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-13, P1] Retroactively run the mandatory independent Fable-xhigh code review (§11.4.125/§11.4.142/§11.4.209) against the substantive diffs the Auto-commit mechanism introduced (start.sh three new functions especially, since they have zero test coverage — see Root Cause 4 / RD2-24) — even though the changes already landed, a post-hoc review closes the governance gap and will surface RD2-24 (GA-27 missing tests) if not already caught. Priority: P1.

BOB-081 — RD2-14: Author CONTINUATION.md Session 15 entry (currently 53 days / 24+ commits behind HEAD)

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-14, P0 — closes GA-04] Author a new CONTINUATION.md Session 15 entry summarizing every substantive commit since 646b295 by theme (Cyrillic/UTF-8 fix wave, tracker/plugin fixes, egress/proxy/Jackett feature wave, the two governance audits, RD2-00 discovery). Update **Last commit:/Last modified:/ Revision:**. GA-04 evidence: still Revision: 19, Last modified: 2026-06-16T23:55:00Z, Session 14 — ~53 days / 24+ commits behind HEAD. Priority: P0.

BOB-082 — RD2-15: Create BOB-064..067 workable items for the four Lava-porting findings (closes GA-05)

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-15, P0 — closes GA-05] Create tracked workable items (BOB-064..067, via the workable-items tool — never raw MD edits) for the four Lava-porting findings, citing implementing commits as evidence, closed as Implemented. GA-05 evidence: grep -in lava|BOB-06[4-7] docs/Issues.md docs/Fixed.md returns zero hits. Priority: P0. NOTE: BOB-064..067 have been created 2026-08-10 as Task/Queued (the four Lava P1..P4 items); the closed-as-Implemented status flip (citing per-finding implementing commits) remains owed under this item.

BOB-083 — RD2-16: Regenerate browser_extension/features/codegraph Status.md + Summary/HTML/PDF siblings

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-16, P1] Regenerate docs/browser_extension/Status.md, docs/features/Status.md, docs/codegraph/Status.md + their Status_Summary.md/HTML/PDF siblings (§11.4.45/§11.4.56). Priority: P1. Composes with RD2-08.

BOB-084 — RD2-17: Reconcile BOB-008 DB/MD body drift via the workable-items tool

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-17, P1] Reconcile BOB-008 DB/MD body drift (RD2-04) via the workable-items tool. Priority: P1. Composes with RD2-04 (evidence source) and RD2-20 (preventive fix).

BOB-085 — RD2-18: Create top-level Boba proxy/merge-service v1.0.0 readiness ledger (closes GA-10)

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-18, P2 — closes GA-10] Create the top-level Boba (proxy/merge-service) v1.0.0 readiness ledger (GA-10) — dedupe with the browser_extension existing one as the template. GA-10 evidence: only docs/RELEASE_READINESS_20260616.html/.md/.pdf (dated point-in-time snapshot) and the extension own ledger exist; no top-level proxy/merge-service ledger created. Priority: P2.

BOB-086 — RD2-19: Fix BOB-009/ BOB-010 evidence_path + backfill item_history for 56 silent closures

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-19, P2] Fix BOB-009/BOB-010 evidence_path values + backfill item_history audit trails for the 56 silent closures where recoverable from git log (RD2-03). Rev-6 note: no workable-items subcommand exists to fix a historical closure evidence_path — see Root Cause 2 in the audit. Priority: P2.

BOB-087 — RD2-20: Wire docs_chain / commit-seam sync hook per §11.4.106(F) so DB writes cannot land without MD mirror

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-20, P0] Wire (or fix) the docs_chain / commit-seam sync hook per §11.4.106(F) so a docs/workable_items.db write can never again land without its MD mirror in the same commit — this is the mechanical fix that prevents Root Cause 2 from recurring, not just a one-time catch-up. Priority: P0.

BOB-088 — RD2-21: Complete/verify README Tracked-Items + Status Documents table row-completeness (GA-07 remainder)

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-21, P1] Complete/verify the README Tracked-Items + Status Documents table row-completeness (GA-07 remaining half). Not independently re-verified this round whether every mandated doc (CONTINUATION.md, Issues.md, Fixed.md, PORTING-FROM-LAVA.md, both new GA/RD2 audit docs, every Status.md/ Status_Summary.md pair) actually has a row. Priority: P1.

BOB-089 — RD2-24: RED-first tests for start.sh reload_python/reload_plugins/ recreate_stack (closes test-half of GA-27)

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-24, P1 TDD — closes test-coverage half of GA-27] Author RED-first tests for start.sh reload_python(), reload_plugins(), recreate_stack() — real executing tests through the actual invocation path (per §11.4.224(A): never a bash -n parse-check alone), asserting exit status + the documented side effects (pycache-clear-then-restart, restart-only, down-then-up) against a live or realistically-mocked container runtime. Verification: tests fail against a stubbed/reverted version of the three functions, pass against current start.sh (§11.4.115). GA-27 evidence: grep -rln reload_python|reload_plugins|recreate_stack tests/ challenges/ returns zero hits — shipped with no test-first coverage, a direct §11.4.224 violation. Priority: P1.

BOB-090 — RD2-25: HelixQA Challenge entry exercising all three start.sh subcommands end-to-end against real compose stack

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-25, P2] Add a HelixQA Challenge entry exercising all three start.sh subcommands (reload_python, reload_plugins, recreate_stack) end-to-end against the real compose stack. Priority: P2.

BOB-091 — RD2-26: Relocate mocked SearchOrchestrator tests to unit/ + author real-service replacements (closes GA-14/15/16)

Status: Queued **Type:** Bug **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-26, P1 — closes GA-14/15/16] For each of tests/integration/test_merge_api.py, tests/e2e/test_full_pipeline.py, tests/contract/test_tracker_stats_contract.py: (a) relocate the existing mocked version into tests/unit/ under an honest name (the coverage is valuable as unit-level route-contract testing — keep it, just where it belongs); (b) author a real-service replacement in the original directory using the repo existing live-stack fixtures (tests/fixtures/compose.py, tests/fixtures/services.py, tests/integration/test_fixtures_bring_up_services.py already prove this pattern works here). Verification: each relocated/replaced file directory-appropriate real-service test actually issues a real HTTP call to a real running service and asserts on the real response — no @patch/monkeypatch targeting SearchOrchestrator anywhere outside tests/unit/. Priority: P1.

BOB-092 — RD2-27: Remove test_live_stack_evidence.py:265 nnmclub SKIP-on-404 fallback + verify live 200 (closes GA-13)

Status: Queued **Type:** Bug **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-27, P1 — closes GA-13] tests/e2e/test_live_stack_evidence.py:265 — with the live stack up, confirm curl :7187/api/v1/auth/nnmclub/status returns 200 (redeploy first if source≠artifact drift persists — see Root Cause 6), remove the skip fallback entirely, let the real assertion run. Priority: P1.

BOB-093 — RD2-28: Live compose bring-up + verify rutracker ReDoS regex bounds deployed to container (closes runtime half of GA-12)

Status: Queued **Type:** Task **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-28, P1 — closes runtime-verification half of GA-12] Bring up the live compose stack (./start.sh), run ./install-plugin.sh, verify grep {0,512} config/qBittorrent/nova3/engines/rutracker.py on the container, and capture timing of a large rutracker result page (<2s per the original RW-06 acceptance criterion). GA-12 status: DONE (source) at plugins/rutracker.py:140 confirmed {0,512}/{0,256}-bounded; runtime unverified (no live container stack running at investigation time). Priority: P1.

BOB-094 — RD2-29: Author tests/stress/test_tracker_fetch_stress_chaos.py with §11.4.85 fault injection

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-29, P2, parallelizable] Author tests/stress/test_tracker_fetch_stress_chaos.py (download-proxy tracker-fetch + cookie-auth path, overlapping the already-fixed SSRF surface) with real §11.4.85 fault injection (process-kill, network-fault, resource-exhaustion — not just concurrency). Verification: each new chaos test is negation-proven (fails without the fault-tolerance code, passes with it), captured evidence per §11.4.5/§11.4.69/§11.4.107. Priority: P2. Composes with GA-20.

BOB-095 — RD2-30: Author tests/stress/test_scheduler_hooks_sse_stress_chaos.py for Go-side triangle

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-30, P2, parallelizable] Author tests/stress/test_scheduler_hooks_sse_stress_chaos.py covering the Go-side internal/service/sse_broker.go + internal/api/hooks.go + internal/api/scheduler_api.go triangle. Verification: each new chaos test is negation-proven (fails without the fault-tolerance code, passes with it), captured evidence per §11.4.5/§11.4.69/§11.4.107. Priority: P2. Composes with GA-20.

BOB-096 — RD2-31: Extend qBitTorrent-go jackett_db_test.go with real process-kill/resource-exhaustion fault injection

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-31, P2] Extend qBitTorrent-go/tests/integration/jackett_db_test.go with real process-kill / resource-exhaustion fault injection (currently concurrency-only per GA-20). Priority: P2.

BOB-097 — RD2-32: Author DDoS-class coverage for exposed download-proxy/merge endpoints (canonical impl of RD2-07)

Status: Queued **Type:** Task **Severity:** Low

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-32, P3] Author DDoS-class coverage (RD2-07) for the exposed download-proxy/merge endpoints. Priority: P3.

BOB-098 — RD2-34: Parametrize 20 hardcoded /Volumes/T7 paths in helixqa banks with PROJECT_ROOT (closes GA-23)

Status: Queued **Type:** Task **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-34, P1 — closes GA-23] submodules/helixqa/banks/*.yaml — parametrize the 20 hardcoded /Volumes/T7/Projects/Boba/… paths with $PROJECT_ROOT (GA-23) — lands in the submodule per*

§11.4.28 decoupling, not boba own tree. GA-23 evidence: grep -rn /Volumes/T7 submodules/helixqa/banks/.yaml returns 20 hits. Priority: P1.

BOB-099 — RD2-36: Fix guard-forbidden-commands.sh substring-match false-positive class + add const033-poweroff-signal-triage carrier to EXCLUDE_PATHS

Status: Queued **Type:** Bug **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-36, P2] Fix guard-forbidden-commands.sh substring-match false-positive class (RD2-01) — word-boundary/token-aware matching, not another EXCLUDE_PATHS band-aid; also add the newly-discovered docs/incidents/2026-08-07-const033-poweroff-signal-triage.md carrier to EXCLUDE_PATHS as an immediate stopgap (GA-24 residual). Priority: P2. Composes with RD2-01, GA-24 residual, and RD2-41 (same class in mutation-marker scan).

BOB-100 — RD2-39: Bump submodules/jackett one commit (canonical impl of RD2-09)

Status: Queued **Type:** Task **Severity:** Low

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-39, P3] Bump submodules/jackett one commit (RD2-09). Priority: P3.

BOB-101 — GA-19/RW-09: Is –profile go parity still a release goal? (gates RW-10..13) — OPERATOR-DECISION

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: OPERATOR must decide whether Go –profile go parity remains a v1.0.0 release goal; gates downstream RW-10..13. WHY: §11.4.66 forbids autonomous decision on release scope; §11.4.122 forbids silent removal of an existing capability. UNBLOCK: [A] operator states YES — parity remains a release goal (schedule

RW-10..13 work) · [B] operator states NO — mark parity Obsolete/ superseded per §11.4.90 (reason=feature-removed with operator citation, §11.4.122) WHO: Operator **Severity:** High

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md GA-19/RW-09, OPERATOR-DECISION] Confirmed unchanged: zero time.Ticker anywhere in qBitTorrent-go, no enricher package, zero exec.Command fan-out. Is –profile go parity still a release goal? Gates RW-10..13. Surfaced only, not auto-executed, per §11.4.66. Priority: OPERATOR-DECISION.

BOB-102 — RW-05: LAN-exposure threat model — bind tunnel 127.0.0.1 or keep 0.0.0.0? — OPERATOR-DECISION

Status: Operator-blocked **Type:** Task **Operator-Block-Details:** WHAT: OPERATOR must decide LAN-exposure posture: bind tunnel to 127.0.0.1 (loopback-only) OR keep 0.0.0.0 (LAN-reachable, relying on post-RD2-22 complete auth coverage). WHY: This is a security-posture policy call the agent cannot make autonomously (§11.4.66 operator-decision, §11.4.101 high-blast-radius reversible-only-if-explicit). UNBLOCK: [A] operator states BIND 127.0.0.1 (loopback-only; agent will change compose/ service binding + regression-test LAN unreachability) · [B] operator states KEEP 0.0.0.0 (rely on §RD2-22 completed auth coverage; agent will add a permanent §11.4.135 LAN-auth regression guard) WHO: Operator **Severity:** Medium

[Backfill from GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RW-05 ungrouped, OPERATOR-DECISION] LAN-exposure threat model — bind the tunnel 127.0.0.1 or keep 0.0.0.0 with the now-complete auth coverage (post Root Cause 3)? Priority: OPERATOR-DECISION.

BOB-104 — §11.4.238 followup: CodeGraph 1.5.0 nested-.gitignore regression challenge

Status: In progress **Type:** Task **Severity:** Medium **Created-By:** Claude

Coverage-escape followup (docs/QA_DISCOVERY_LEDGER.md, BOB-075 agent-code-reading finding, commit e6162f7): CodeGraph 1.5.0 (up from documented 0.9.9) walked into nested-.gitignore-excluded frontend/node_modules and extension/ node_modules (32,260 files / 514,456 nodes vs the 2026-06-06 baseline of 509 files / 8,906 nodes) instead of honoring frontend/.gitignore + extension/.gitignore per git check-ignore -v. Author a challenge (challenges/scripts/codegraph_gitignore_honor_challenge.sh or equivalent, e.g. wrapping 'codegraph doctor –sniff-gitignore-honor' if that subcommand exists, else a real re-index + file-count assertion) with §11.4.115 RED_MODE polarity: RED_MODE=1 reproduces the blowup against the live nested-.gitignore tree, RED_MODE=0 asserts the resync stays within the documented baseline order of magnitude. Full evidence: docs/codegraph/Status.md lines 91-118. UPSTREAM FILED 2026-08-18: real upstream repo is github.com/colbymchenry/codegraph (npm package @colbymchenry/codegraph, confirmed via package.json + gh repo view; NOT vasic-digital/codegraph, correcting this ledger's original assumption). Issue: https://github.com/colbymchenry/codegraph/issues/1567 (full reproduction evidence: git check-ignore -v re-verified first-hand; two bounded synthetic repro attempts up to 63 nested .gitignore files / 960 files against the actual installed v1.5.0 binary did NOT reproduce the blowup in isolation – honest negative result, root cause not pinned to a specific line in either scanning implementation). Draft PR (diagnostic only, not a behavioral fix): https://github.com/colbymchenry/codegraph/pull/1568 – adds a logDebug() call so a future report can confirm which of the two independent gitignore-respecting code paths (git-ls-files-based vs filesystem-walk fallback) actually ran. Both PRs/issue filed via SSH (Hard Stop #2) from a fork at github.com/milos85vasic/codegraph. Status: In-progress pending upstream maintainer triage/merge – boba-side closure of this item still needs the RED/GREEN challenge script (§11.4.115) authored per the original scope, independent of upstream's response.

BOB-105 — §11.4.238 followup: mechanical §11.4.227(B) anchor-block-integrity check

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

Coverage-escape followup (docs/QA_DISCOVERY_LEDGER.md, meta-defect found during this session's §11.4.140/141 collision resolution, boba commit 136a22c + constitution commits 5ed8c80..e5f2891): §11.4.227(B) anchor-block-integrity (exactly-once heading per anchor per governance file, byte-identical lockstep across mirrors, no anchor-number collision) is fully specified in constitution prose but its own named gate CM-ANCHOR-BLOCK-INTEGRITY is explicitly 'gate-code = separate work item, NOT claimed shipped' — the §11.4.140/§11.4.141 double-mandate collision this session's Phase 0 resolved was verified entirely BY HAND (md5 of moved anchor bodies, manual grep counts pasted into a commit message), not by a runnable script. Propose + author a challenge (in this project's challenges/scripts/, since this project cannot edit the constitution submodule's own gate-code from this ticket) that greps every governance file (CLAUDE.md/AGENTS.md/QWEN.md/GEMINI.md/Constitution.md, wherever this project's constitution submodule checkout exposes them) for '### §11.4.NNN' heading occurrences and FAILs on >1-per-file-per-anchor or on two DIFFERENT anchor bodies sharing one NNN, per §11.4.227(B). Do NOT touch constitution-owned files while implementing this — the check reads them read-only.

BOB-106 — §11.4.238 followup: §11.4.84 quiescence-check helper for the unattributed auto-commit path

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

Coverage-escape followup (docs/QA_DISCOVERY_LEDGER.md RD2-00/BOB-068 entry, docs/GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-00): 20 bare 'Auto-commit' commits exist in this repo's git history (confirmed via git log –oneline –all –grep, e.g. 54e313f/9c8f684/743097a/de9270b/1c36777/41179c2/7c529ca) with no ATM-NNN reference and no TDD trail, landing via ordinary git pull fast-forward from a second session/host with push access to the same remotes (mismatched commit timezone vs the investigating host, per RD2-00 Update). §11.4.84 working-tree-quiescence has no mechanical guard on this path — no gate flags a commit reaching main with a bare/templated message and no ticket citation. Author a §11.4.84 quiescence-check helper (e.g. challenges/scripts/no_unattributed_autocommit_challenge.sh) that scans the commit range since the last known-good release tag and FAILs on any commit message matching a closed bare/templated pattern (e.g. ^Auto-commit$, ^sync:) with no ATM-NNN or task/PR reference, wired into scripts/pre_build_verification.sh or the §11.4.234 commit-push-all.sh entrypoint. BOB-068 (RD2-00) remains the tracking item for identifying/stopping the source; this item is specifically the new automated CHECK.

BOB-107 — §11.4.238 followup: pre-dispatch existence check for subagent task-brief source inputs

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

Coverage-escape followup (docs/QA_DISCOVERY_LEDGER.md SCRATCH-LOSS-2026-08-18 entry, evidence: .superpowers/sdd/task-phase1a-report.md line 21): the Phase 1a subagent (a1cc331d) discovered at task start that 5 source files its brief named as required reads (curriculum_amendment_plan_v1.md, ai_curriculum_modules_27_35_extracted.md, and 3 curriculum_analysis_modules_*.md gap analyses) were absent from the session scratchpad — root cause per that subagent's own investigation: the prior producer subagent (ae59171f) hit its session rate limit (a §11.4.147(e) API-quota crash) before writing them. curriculum_amendment_plan_v1.md remains absent from the live scratchpad as of this session's re-check. No mechanical check verifies a task brief's declared input files exist and are non-empty before the downstream consumer subagent is dispatched. Author a pre-dispatch precondition helper (project-side orchestration tooling, e.g. a small script the conductor runs before Task/Agent dispatch when a brief names required input paths) that fails closed with an actionable 'missing input, respawn the producer' message rather than silently letting a downstream agent proceed on absent evidence and fabricate content unsupported by its named sources.

BOB-109 — BOB-074 followup: scaling-class test coverage absent from mandated test-type matrix

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

docs/testing/test_type_matrix.md's §11.4.27 test-type audit found zero scaling-class coverage anywhere in the tree (no scaling-tagged directory, test file, or HelixQA bank distinguishes growing-dataset/tracker-count/concurrent-user scale-out from stress-under-burst). Scope at least one scaling dimension, e.g. tracker-count scale-out in merge search against challenges/helixqa-banks/boba-services.yaml's tracker set, or the qbittorrent-proxy-go –profile go swap, with a real measured baseline.

BOB-110 — BOB-074 followup: UX-class test coverage (accessibility/usability) absent

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

docs/testing/test_type_matrix.md's §11.4.27 test-type audit found UI functional coverage (Vitest + Playwright) but nothing framed around usability/accessibility/UX outcomes specifically. Scope an axe-core or equivalent accessibility pass over the Angular frontend, covering WCAG checks, keyboard-nav coverage, and screen-reader labeling.

BOB-111 — BOB-074 followup: configure real rate limiting for boba's 3 public HTTP endpoints

Status: In progress **Type:** Task **Severity:** High **Created-By:** Claude

Source inspection across the whole stack (2026-08-18) verified no rate-limit mechanism exists for :7185 (qBittorrent WebUI proxy), :7187 (merge search service), or :7189 (boba-jackett) – no slowapi/limiter/throttle import in download-proxy/src/, no rate-limit middleware in qBitTorrent-go/internal/middleware/ (only cors.go + logging.go) or internal/jackettapi/ (only auth/cors middleware tests, no rate middleware), and no nginx/reverse-proxy service in docker-compose.yml. Candidate remediations documented in docs/testing/ddos_resilience.md: nginx-in-container reverse proxy with limit_req_zone (most portable, adds a container); a FastAPI slowapi dependency for the merge-search service; a Gin rate-limit middleware for boba-jackett following the existing internal/middleware/ pattern. qBittorrent's own WebUI bandwidth-shaping settings were NOT verified to cover request-rate (only bandwidth) – do not assume they close this gap without checking.

BOB-113 — BOB-074 followup: add wrk to dev tooling for DDoS/load challenges

Status: Queued **Type:** Task **Severity:** Low **Created-By:** Claude

wrk is not installed on the development host; ApacheBench (ab) is, and was used as the primary DDoS-resilience-challenge load tool per the task brief's 'wrk or ab' fallback path, with a curl-parallelloop as the actual per-status-code assertion mechanism (more precise than ab's aggregate summary) and ab's own output pasted as supplementary real-tool evidence. Install wrk (e.g. via the project's setup.sh or a documented apt/build-from-source recipe) so future load/DDoS challenges have a modern HTTP benchmarking tool with latency-percentile histograms available out of the box, matching the brief's stated primary preference.

BOB-114 — BOB-074 followup: self-validation golden-bad fixture for the rate-limit detector

Status: Queued **Type:** Task **Severity:** Medium **Created-By:** Claude

challenges/scripts/ddos_resilience_challenge.sh's –self-validate mode currently only ships a golden-bad fixture for the crash-resistance detector (per §11.4.107(10)/§11.4.201). The rate-limiting detector has no matching golden-bad fixture proving it would actually FAIL a synthetic no-rate-limit-enforced artifact, so an unvalidated rate-limit detector could silently pass a broken/absent rate-limit deployment. Add a synthetic fixture (e.g. a stub server that never returns 429/503 under burst) and assert the detector correctly reports the absence, closing the self-validation gap for this detector class.

BOB-117 — rutracker login diag still uses forbidden §11.4.6 'likely' vocabulary + wrong error_type (unfixed sibling of nnmclub fix)

Status: Queued **Type:** Bug **Severity:** High **Created-By:** AI

search.py:1329 (_search_rutracker password-login path) still classifies a no-session-cookie login result as error_type=auth_failure with message 'rutracker login returned no session cookie — likely CAPTCHA wall or credential failure'. The adjacent code comment (lines 1244-1248, same function) already states as FACT that 'rutracker gates login.php behind a CAPTCHA when logins spike… the CAPTCHA is the real blocker' — yet the error classification still hedges with forbidden §11.4.6 vocabulary (likely) and reports the WRONG error_type (auth_failure instead of upstream_captcha), unlike the exact same class of bug already fixed for nnmclub per docs/qa/nnmclub-login-diagnosis-20260616.md (which replaced 'likely credential failure' with a FACT-based upstream_captcha classification + explanatory comment explicitly citing 'This is NOT likely a credential failure').

The rutracker code path could disambiguate the same way nnmclub's fix does (or by reading the login response body for a captcha marker, mirroring the existing pattern used ~20 lines later in the SAME function for the search response) but currently discards that opportunity and ships a guess. Found during a §11.4.6 bluff audit (docs/qa/task-bluff-audit/). Fix direction: apply the same upstream_captcha reclassification + FACT-based message used for nnmclub, or add a captcha-marker check on the login response before falling back to a hedged message.

BOB-118 — README.md python-tests badge claims 585 passing; pytest –collect-only measures 5235 (9x stale/wrong)

Status: Queued **Type:** Bug **Severity:** High **Created-By:** AI

README.md line 28 ships a green/success-colored shields.io badge 'python tests-585 passing', and line 333 repeats '585+ tests'. A real, non-destructive verification this session (timeout 90 nice -n 19 ionice -c 3 .venv/bin/python -m pytest tests/ –collect-only -q) collected 5235 tests, not 585 — roughly 9x higher, so the badge is either wildly stale or was never machine-derived (violates §11.4.259 CM-BADGE-MACHINE-DERIVED-SOURCE + §11.4.6 no-guessing). The README's own §11.4.259 provenance footnote (lines 40-57) explicitly discloses that the tests/vitest/plugins/merge/scan/license badges were 'carried forward unverified this session (out of this task's scope)' and directs readers to docs/TESTING.md 'for the current authoritative per-suite counts' — but docs/TESTING.md contains NO occurrence of '585' or '182' anywhere, so the cited authoritative source cannot actually corroborate the badge. The frontend vitest badge (182 passing) is also suspect: a source-level grep of it()/test() calls in frontend/src/**/*.spec.ts counts ~371, roughly 2x the claimed 182 (lower-confidence proxy than the pytest collect-only count, but consistent with the same staleness pattern). Found during a §11.4.6 bluff audit (docs/qa/task-bluff-audit/). Fix direction: wire a real badge-computer (README already flags this as an owed §11.4.197 gate-code item) that regenerates the badge from a real collect/run count, and add the actual per-suite counts to docs/TESTING.md so the cited authoritative source is real.

BOB-120 — 3rd forced-logout incident 2026-08-18 23:45:49 — SIGKILL user@1000 + preventive-timer-inside-user-slice architectural gap

Status: Queued **Type:** Bug **Severity:** Critical **Created-By:** AI

user@1000.service SIGKILLed at 23:45:49 (2h55m after BOB-116's 20:50:59 kill, same session lineage), cascade-killing gnome-shell/Xwayland/ssh/dbus/gnome-keyring/all MCP servers/ all boba containers' host processes (73 Killing-process lines captured). SIGKILL source UNCONFIRMED per §11.4.6 (kernel OOM clean, systemd-oomd no entries, no lid events, HandleLidSwitch=ignore unchanged) — same honest boundary as BOB-116. NEW architectural finding: boba-resource-pressure-check.timer (BOB-116/task-77's preventive remediation) is a systemd –user unit hosted inside user@1000.service itself, so it dies in the SAME SIGKILL cascade it exists to detect precursors to; captured fire history shows last completed run finished 22:57:58, next due no earlier than 23:52:58, so the timer was not overdue at the 23:45:49 kill — it structurally cannot outlive the scope it monitors, meaning no in-scope self-monitor can ever catch a kill that takes it down with it. Contributing observations: qbittorrent-proxy health-probe ConnectionResetError cascade every ~30s through 23:38-23:45 (last line 23:45:44.737, 5s before kill, correlation only, not established causal); all four boba-stack containers (qbittorrent/jackett/boba-jackett/qbittorrent-proxy) recreated with CreatedAt=23:45:52, 3s after the kill, before any human session existed, confirming container supervision survives independently of user@1000.service. Fix direction: an out-of-user-scope watchdog (root-level systemd timer or a –user unit escaped from the session scope) is required to close this class of gap by construction. Evidence: docs/qa/BOB-120/ {journalctl_23-42_to_23-46.log,timer_fire_history.log,qbittorrent_proxy_socketserver_traces.log}. Full writeup: docs/incidents/ 2026-08-18-3rd-forced-logout.md. Cross-refs: BOB-116 (docs/ incidents/2026-08-18-perceived-forced-logout-2nd.md), §12.12 (RLIMIT_NPROC awareness anchor from the 2026-07-07 1st incident), task #77 (the timer this incident found the limitation of), task #79 (SIGKILL-source attribution, still open after 3 incidents). Left Queued — closing requires the actual out-of-scope-watchdog architectural fix (a documentation-only close would be a §11.4.238 coverage-escape bluff).

BOB-121 — External watchdog for the forced-logout architectural gap (task #85, incident #3)

Status: Queued **Type:** Task **Severity:** Important **Created-By:** Claude

Phase 1 design-only proposal: the BOB-116/task-77 resource-pressure preventive systemd –user timer runs inside user@1000.service, the exact pool it monitors, so it cannot fire when that pool dies (proven by incident #3, docs/qa/BOB-120/, 22:42+22:57 fires then blocked 23:45:49-23:49:00). Recommends Option B (user crontab reusing pre-existing crond.service in system.slice, no new root service) kept alongside the existing timer, with Option A (new root-owned systemd unit) as escalation path if Phase 1.5 live cron/cgroup verification is adverse. Proposal: docs/proposals/external-watchdog-for-forced-logout-architectural-gap.md. Operator decision required per §11.4.66 before any implementation – NOT implemented in this task.

BOB-129 — Potential production slowapi/starlette defect flagged by Task 105 subagent

Status: Queued **Type:** Bug **Severity:** Medium

Task #105 subagent (fixing 9 slowapi test failures) reported honestly that the same slowapi/starlette incompatibility likely hits the production /search and /search/stream endpoints under real HTTP traffic — evidence: the FastAPI TestClient (which goes through the full ASGI middleware stack like real requests do) reproduces the same isinstance() failure pattern the 9 test failures exhibited. Not yet reproduced against the running boba stack because the qbittorrent-proxy container currently exposes no host ports (running on gluetun network stack). Recommended investigation: (1) confirm defect by triggering /search kickoff through gluetun network stack, (2) if reproduced, determine whether the fix belongs in production code (adding response: Response params) or a version pin (slowapi vs starlette compat) or a middleware refactor. §11.4.238 discovery-channel escape prevention: manual QA must NOT be the discoverer. §11.4.108 Layer 3 verification: needed on a clean deployment before any release.

BOB-131 — qbittorrent-proxy podman conmon crash — pre-existing, surfaced during BOB-129 investigation

Status: Queued **Type:** Bug **Severity:** Medium

BOB-129 subagent found qbittorrent-proxy container DEAD mid-investigation (podman conmon crash). Recovery via ./start.sh –no-build worked. Pre-existing, unrelated to BOB-129 slowapi work but surfaced by it. Investigation needed: (a) how long was container dead before discovery? (b) what triggered the conmon crash? (c) is there a §11.4.144 always-follow / §11.4.128 always-record signal we should add to detect this class earlier? Post-recovery: container now unhealthy (see BOB-132). §11.4.238 discovery-channel escape: was originally found by a subagent investigating something else, not by dedicated container health monitoring.

BOB-135 — Test isolation: test_list_hooks_after_create fails in bulk suite (Permission denied /config)

Status: Queued **Type:** Bug **Severity:** Low

Task #109 subagent found: tests/unit/test_merge_api_route_contracts.py::TestHooksEndpoint::test_list_hooks_after_create fails when run in bulk suite order with 'ERROR api.hooks:hooks.py:102 Failed to save hooks: [Errno 13] Permission denied: /config'. Passes in isolation (2.02s clean). Root cause: full-suite ordering pollution — some earlier test leaves state that makes hooks try to write to /config (which the test env doesn't own). Pre-existing, unrelated to BOB-126/BOB-129 chain. Fix strategy: identify the polluting test, add teardown or use a proper tempdir fixture for hooks storage in the offending test.